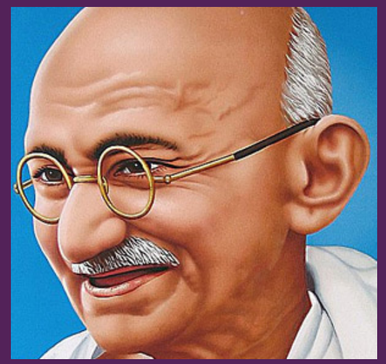


Explorartory Data Analysis with MySQL & Power BI on National Rural Employment Guarantee Act (NREGA) Data

Presented by:
Ranjitha H S
Data Analyst Intern

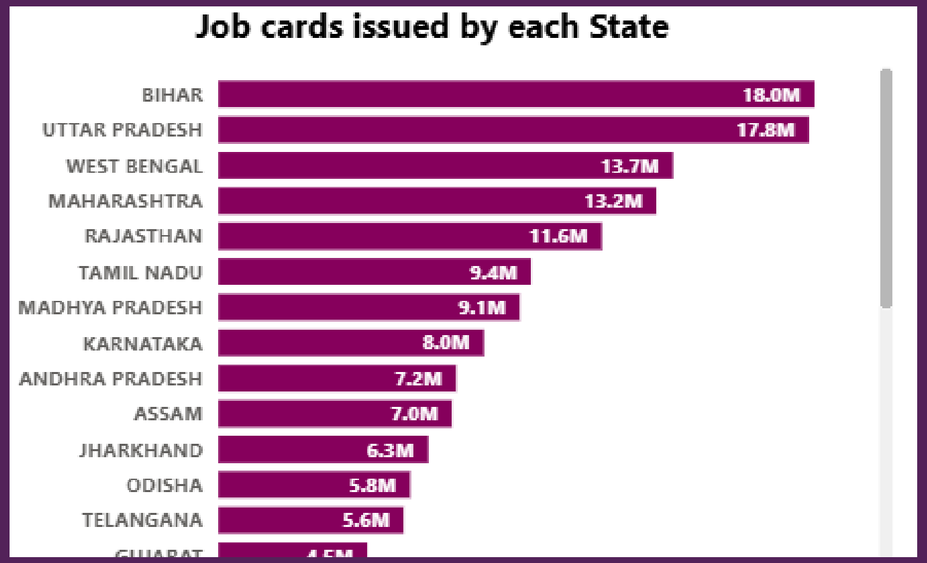
Job Cards & Workers

QUESTION - 1

How many Job cards issued in each State?

```sql
SELECT
    state_name,
    SUM(total_jobcards_issued) AS Total_job_cards
FROM
    mgnrega_data
GROUP BY
    state_name
ORDER BY Total_job_cards desc;
```

	State_Name	Total_job_cards_issued
1	BIHAR	17987345
2	UTTAR PRADESH	17810489
3	WEST BENGAL	13718985
4	MAHARASHTRA	13216917
5	RAJASTHAN	11562058
6	TAMIL NADU	9433367
7	MADHYA PRADESH	9088580
8	KARNATAKA	8012120
9	ANDHRA PRADESH	7170051
10	ASSAM	7046513
11	JHARKHAND	6325674



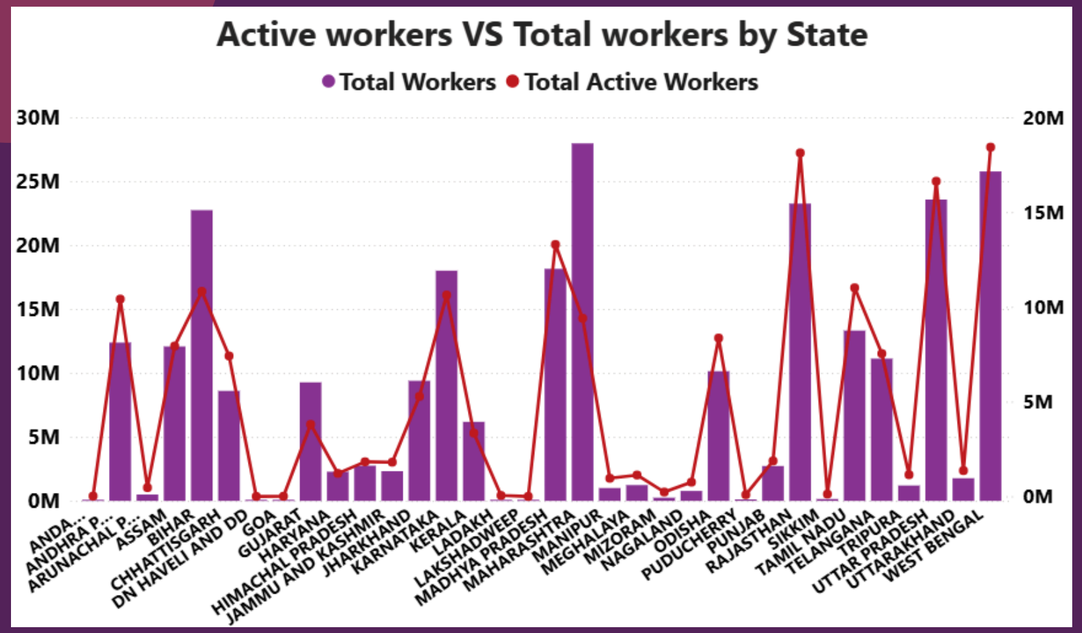
Job Cards & Workers

QUESTION - 2

Active workers VS Total workers by State

```sql
SELECT
    state_name,
    SUM(total_workers) AS total_workers,
    SUM(total_active_workers) AS total_active_workers,
    CEILING(
        (SUM(total_active_workers) * 1.0 / SUM(total_workers)) *
    ) AS percentage_of_active_workers
FROM mgnrega_data
GROUP BY state_name
Order By state_name;
```

state_name	total_workers	total_active_workers	percentage_of_active_workers
ANDAMAN AND NICOBAR	56857	18947	34
ANDHRA PRADESH	12380375	10412371	85
ARUNACHAL PRADESH	497643	464771	94
ASSAM	12077132	7934595	66
BIHAR	22743075	10818160	48
CHHATTISGARH	8603102	7412397	87
DN HAVELI AND DD	33136	2	1
GOA	50713	10091	20
GUJARAT	9270975	3803098	42
HARYANA	2281305	1216597	54
HIMACHAL PRADESH	2751680	1828473	67
JAMMU AND KASHMIR	2323042	1808452	78
JHARKHAND	9390515	5280500	57
KARNATAKA	18006305	10627340	60



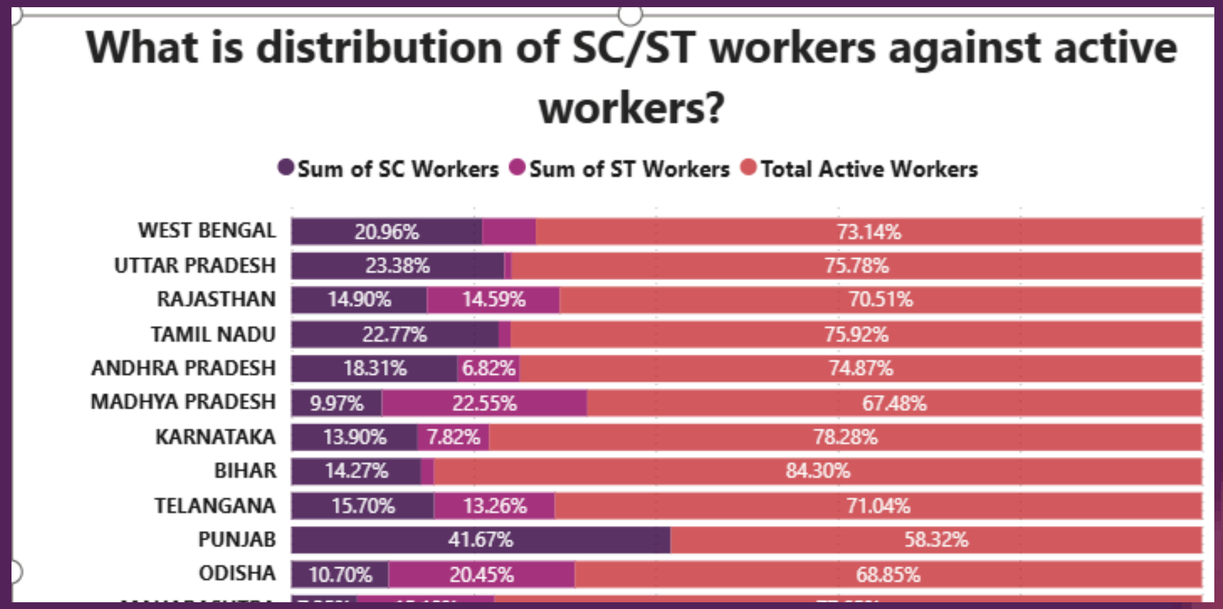
Job Cards & Workers

QUESTION - 3

What is distribution of SC/ST workers against active workers?

```sql
SELECT
    state_name,
    CAST(ROUND((SUM(sc_active_workers) * 100.0)/ SUM(total_active_workers),2) AS FLOAT)
    AS percentage_of_sc_active_workers,
    CAST(ROUND((SUM(st_active_workers) * 100.0)/ SUM(total_active_workers), 2)AS FLOAT)
    AS percentage_of_st_active_workers
FROM mgnrega_data
GROUP BY state_name
ORDER BY state_name DESC;
```

state_name	percentage_of_sc_active_workers	percentage_of_st_active_workers
WEST BENGAL	28.65	8.07
UTTARAKHAND	17.59	3.97
UTTAR PRADESH	30.85	1.1
TRIPURA	17.38	41.28
TELANGANA	22.11	18.66
TAMIL NADU	29.99	1.72
SIKKIM	5.24	41.35
RAJASTHAN	21.13	20.69
PUNJAB	71.45	0.03
PUDUCHERRY	30.05	0.26
ODISHA	15.54	29.7



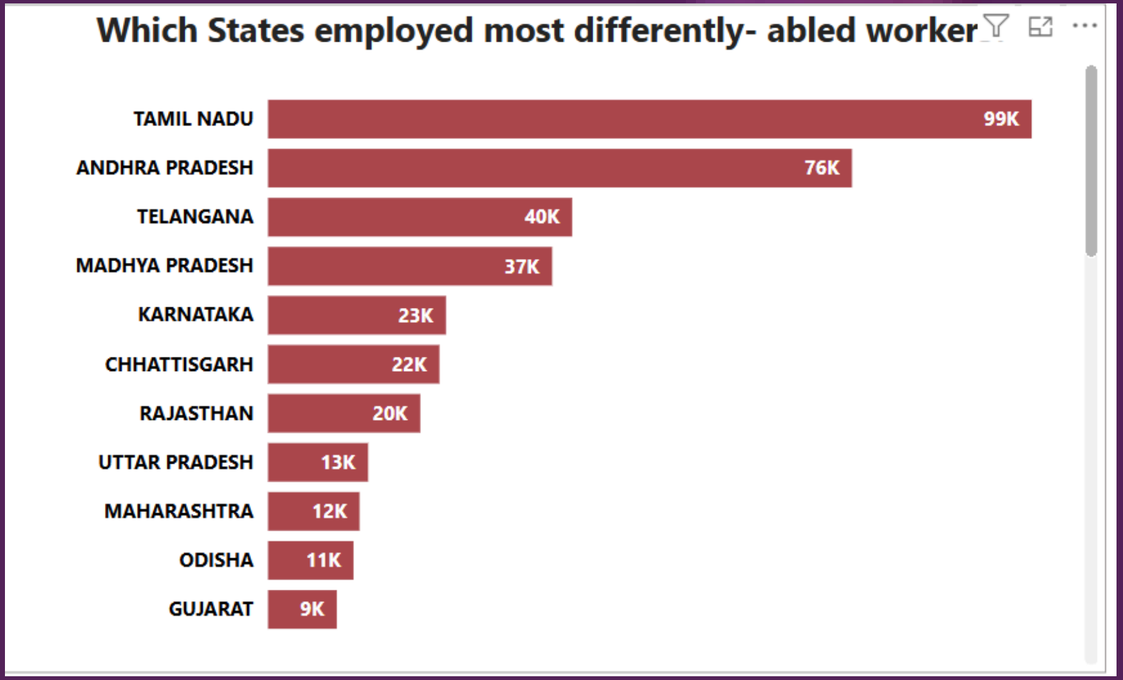
Job Cards & Workers

QUESTION - 4

Which States employed most differently- abled workers?

```sql
SELECT
    state_name,
    SUM(differently_abled_persons_worked) AS total_differently_abled_workers
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_differently_abled_workers DESC;
```

state_name	total_differently_abled_workers
TAMIL NADU	99285
ANDHRA PRADESH	75917
TELANGANA	39535
MADHYA PRADESH	36937
KARNATAKA	23132
CHHATTISGARH	22280
RAJASTHAN	19790
UTTAR PRADESH	13008
MAHARASHTRA	11893
ODISHA	11109
GUJARAT	8939



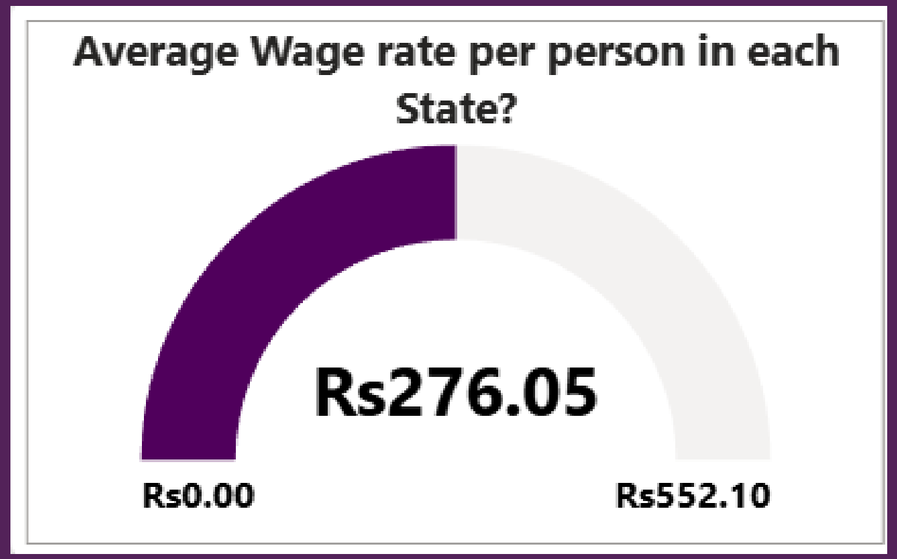
Works & Employment

QUESTION - 5

What is Average wage rate per day for all states??

```sql
SELECT
    CAST(ROUND(AVG(avg_wage_per_day), 2) AS FLOAT)
        AS average_wage_rate_per_day
FROM mgnrega_data;
```

average_wage_rate_per_day
276.05

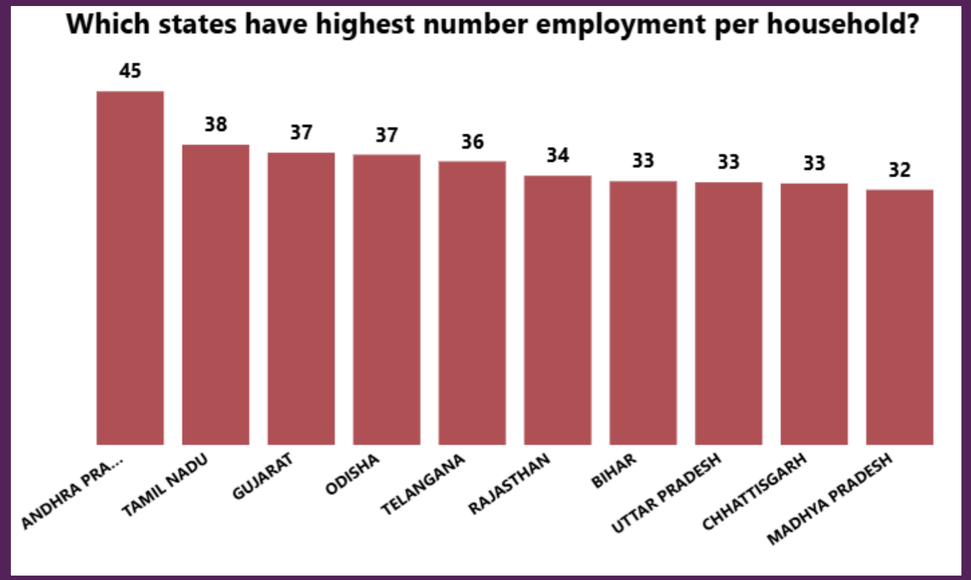


QUESTION - 6

What is the average number of days of employment provided per household in each state?

```sql
SELECT
    state_name,
    CEILING(AVG(avg_days_employment_per_household))
        AS avg_days_employment_per_household_per_state
FROM mgnrega_data
GROUP BY state_name
ORDER BY avg_days_employment_per_household_per_state DESC;
```

	state_name	avg_days_employment_per_household_per_state
1	ANDHRA PRADESH	45
2	MIZORAM	38
3	TAMIL NADU	38
4	TRIPURA	37
5	NAGALAND	37
6	ODISHA	37
7	GUJARAT	37
8	JHARKHAND	36
9	TELANGANA	36



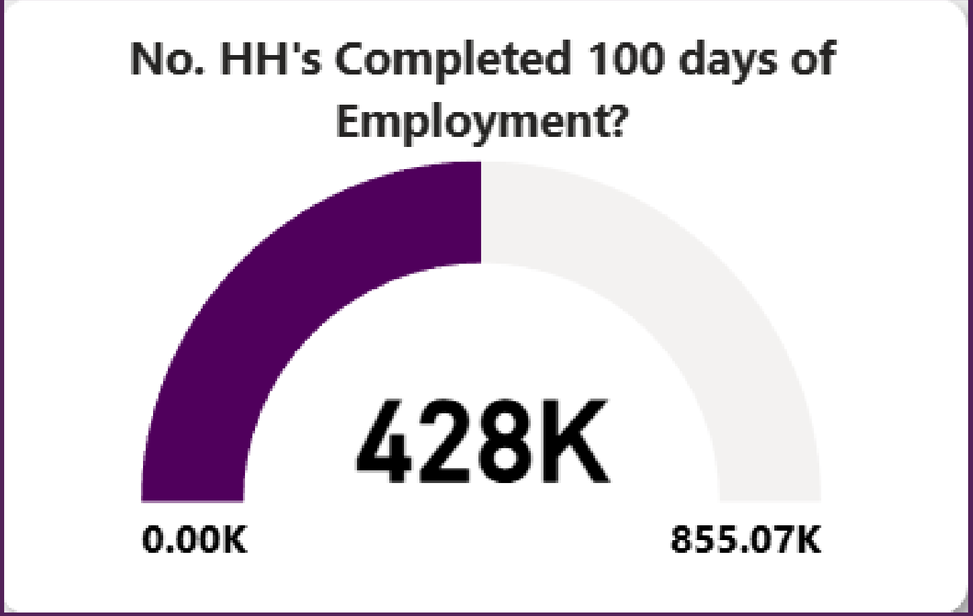
Works & Employment

QUESTION- 7

How many households have completed 100 days of wage employment in each state?

```sql
SELECT
    state_name,
    SUM(households_completed_100_days)
        AS households_completed_100_days
FROM mgnrega_data
GROUP BY state_name
ORDER BY households_completed_100_days DESC;
```

state_name	households_completed_100_days
ANDHRA PRADESH	119939
ODISHA	77987
MAHARASHTRA	49534
UTTAR PRADESH	33818
CHHATTISGARH	23501
TAMIL NADU	19730
RAJASTHAN	17926
JHARKHAND	14587
KARNATAKA	13811
MADHYA PRADESH	12436



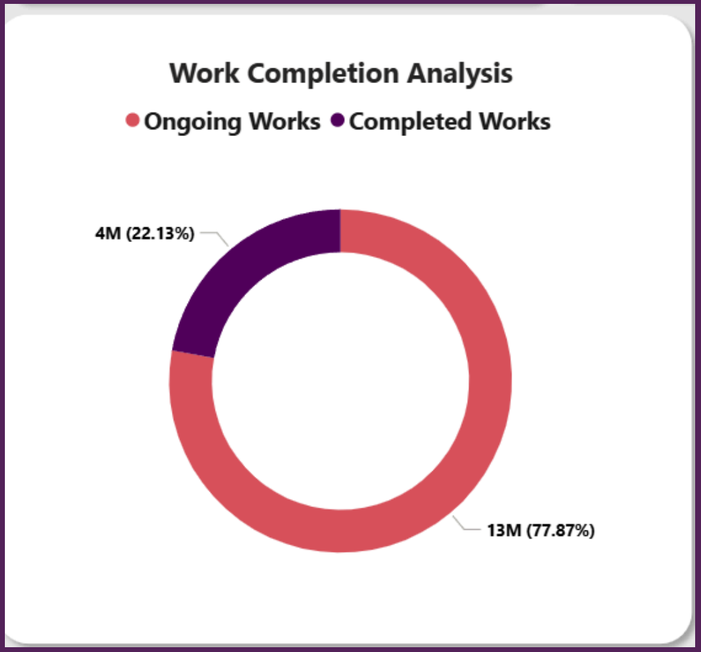
Works & Employment

QUESTION - 8

Work Completion Analysis

```sql
SELECT
    SUM(total_works_taken_up) AS total_works_taken_up,
    SUM(ongoing_works) AS total_ongoing_works,
    SUM(completed_works) AS total_completed_works
FROM mgnrega_data;
```

total_works_taken_up	total_ongoing_works	total_completed_works
16765863	13055054	3710809



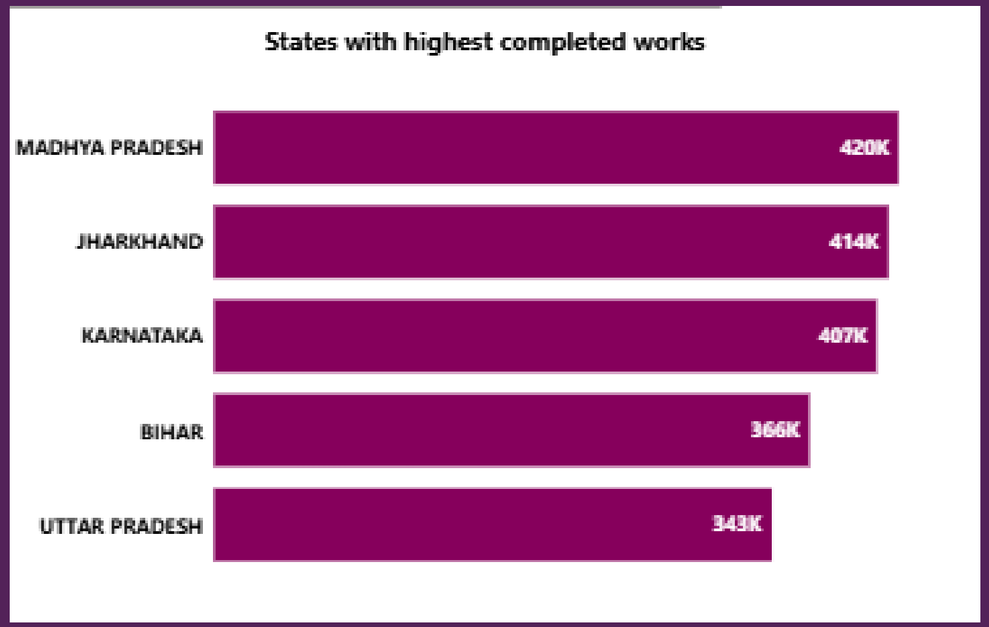
Works & Employment

QUESTION - 9

Which states have the highest number of completed works?

```sql
SELECT
    state_name,
    SUM(completed_works) AS total_completed_works
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_completed_works DESC;
```

state_name	total_completed_works
MADHYA PRADESH	420359
JHARKHAND	414172
KARNATAKA	407120
BIHAR	365721
UTTAR PRADESH	342531



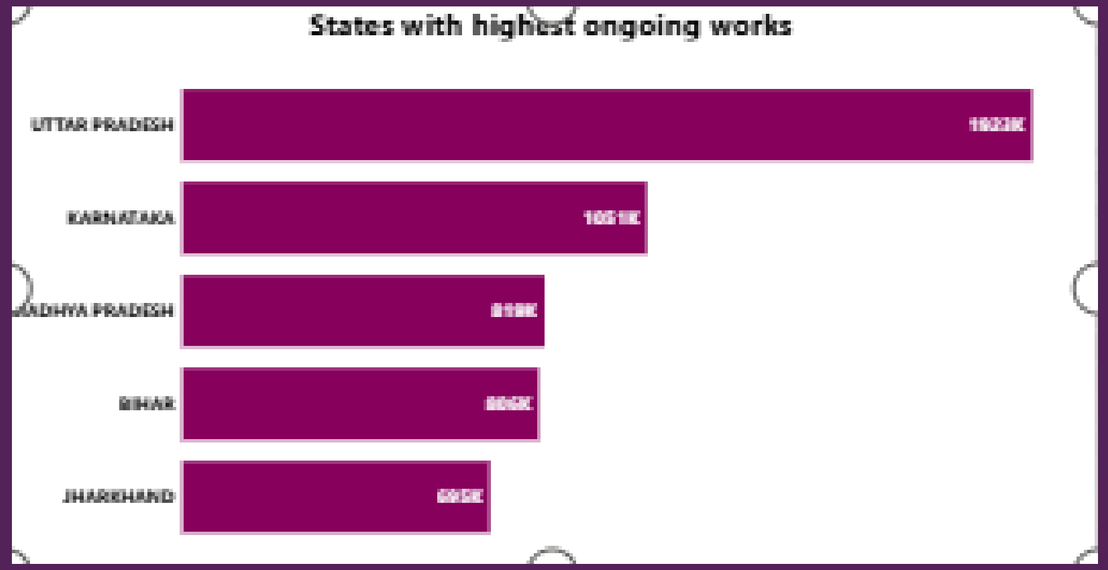
Works & Employment

QUESTION - 10

Which states have the highest number of Ongoing works?

```sql
SELECT TOP 5
    state_name,
    SUM(ongoing_works) AS total_ongoing_works
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_ongoing_works DESC;
```

state_name	total_ongoing_works
UTTAR PRADESH	1923030
WEST BENGAL	1130744
KARNATAKA	1050889
ANDHRA PRADESH	1034159
ASSAM	898785

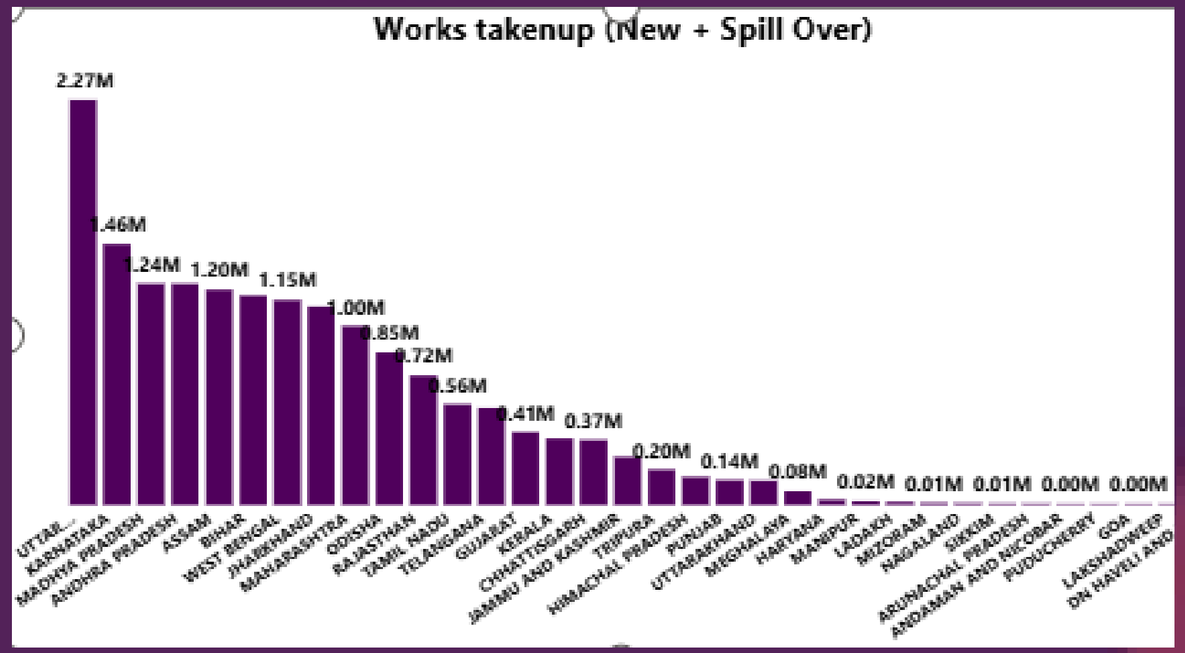


QUESTION - 11

How many total works (new and spillover) have been taken up in each state?

```sql
SELECT TOP 5
    state_name,
    SUM(total_works_taken_up) AS total_works_taken_up
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_works_taken_up DESC;
```

state_name	total_works_taken_up
UTTAR PRADESH	2265561
KARNATAKA	1458009
MADHYA PRADESH	1239576
ANDHRA PRADESH	1239441
ASSAM	1203920



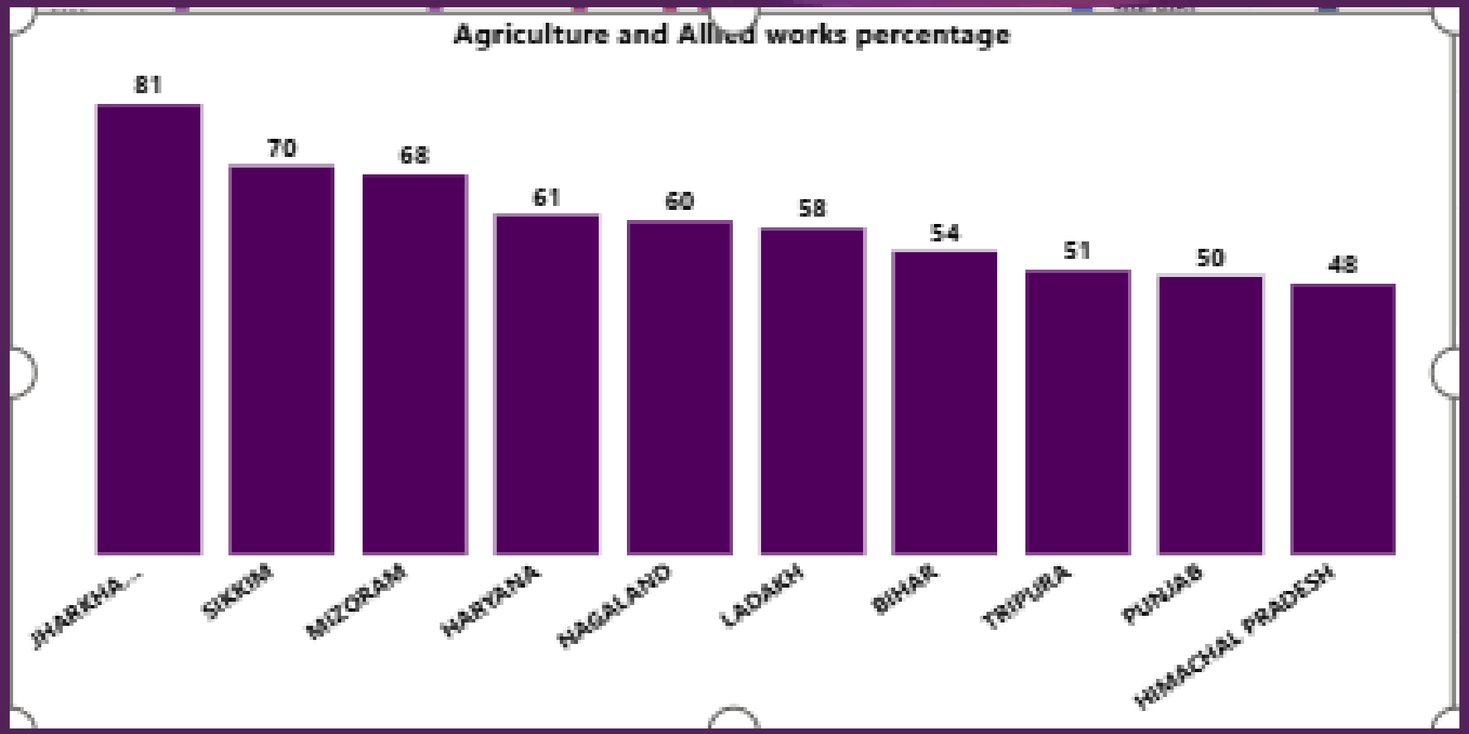
Works & Employment

QUESTION - 12

State-wise Agriculture & Allied workers distribution across States

```sql
select
    state_name,
    cast(round(avg(pct_agri_allied_expenditure),2 ) as float) avg_pct_agri_allied_expenditur
from mgnrega_data
group by state_name
order by avg_pct_agri_allied_expenditure desc;
```

state_name	avg_pct_agri_allied_expenditure
JHARKHAND	80.64
SIKKIM	69.66
MIZORAM	68.08
HARYANA	60.87
NAGALAND	59.72
LADAKH	58.45
BIHAR	54.38
TRIPURA	50.86
PUNJAB	49.91
HIMACHAL PRADESH	48.4
MAHARASHTRA	47.72
UTTARAKHAND	47.08
ODISHA	46.04
ANDHRA PRADESH	43.31
ARUNACHAL PRADESH	41.02



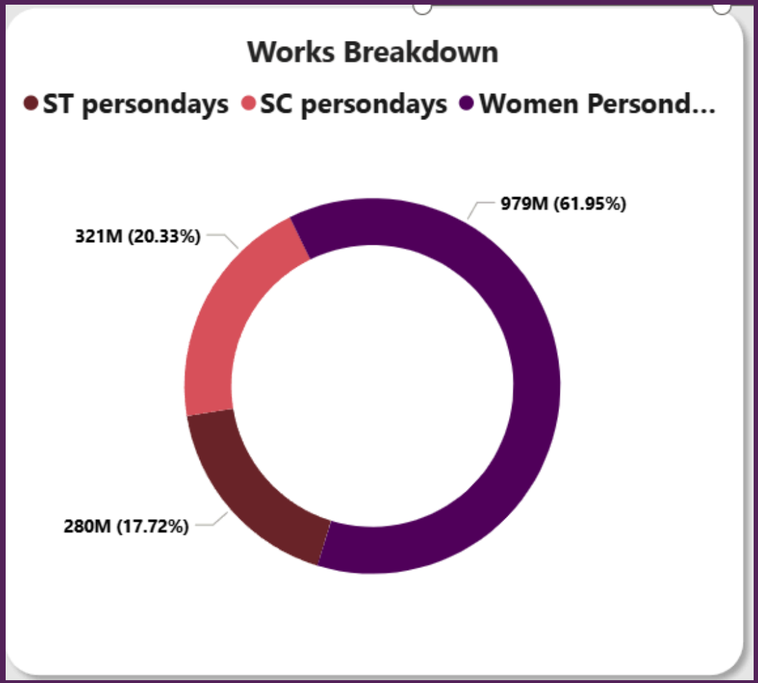
Works & Employent Analysis

QUESTION - 13

Workdays between Active social groups select sum(sc_persondays)

```
SELECT
    SUM(sc_persondays) AS sc_workdays,
    SUM(st_persondays) AS st_workdays,
    SUM(women_persondays) AS women_workdays
FROM mgnrega_data;
```

sc_workdays	st_workdays	women_workdays
321202295	279960190	978758223



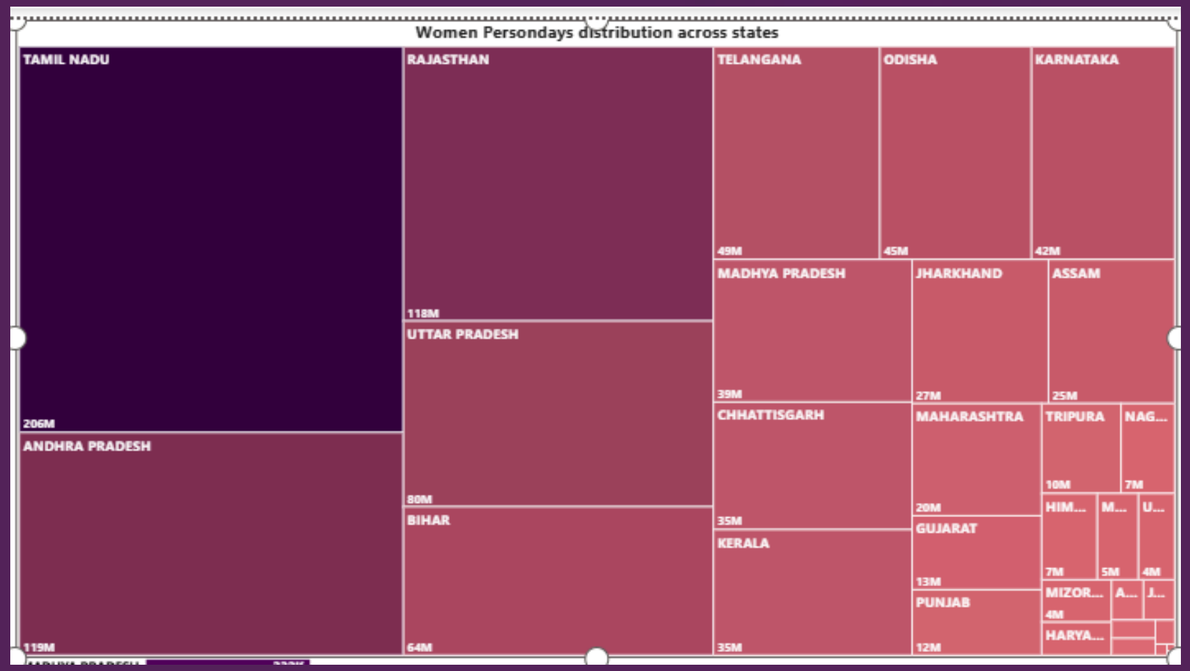
Works & Employent Analysis

QUESTION - 14

How does the number of women persondays vary across states?

```sql
SELECT
    state_name,
    SUM(women_persondays) AS total_women_persondays
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_women_persondays DESC;
```

state_name	total_women_persondays
TAMIL NADU	206283054
ANDHRA PRADESH	119060226
RAJASTHAN	118387733
UTTAR PRADESH	80193649
BIHAR	64085920
TELANGANA	49318998
ODISHA	44925151
KARNATAKA	42387404
MADHYA PRADESH	39445322



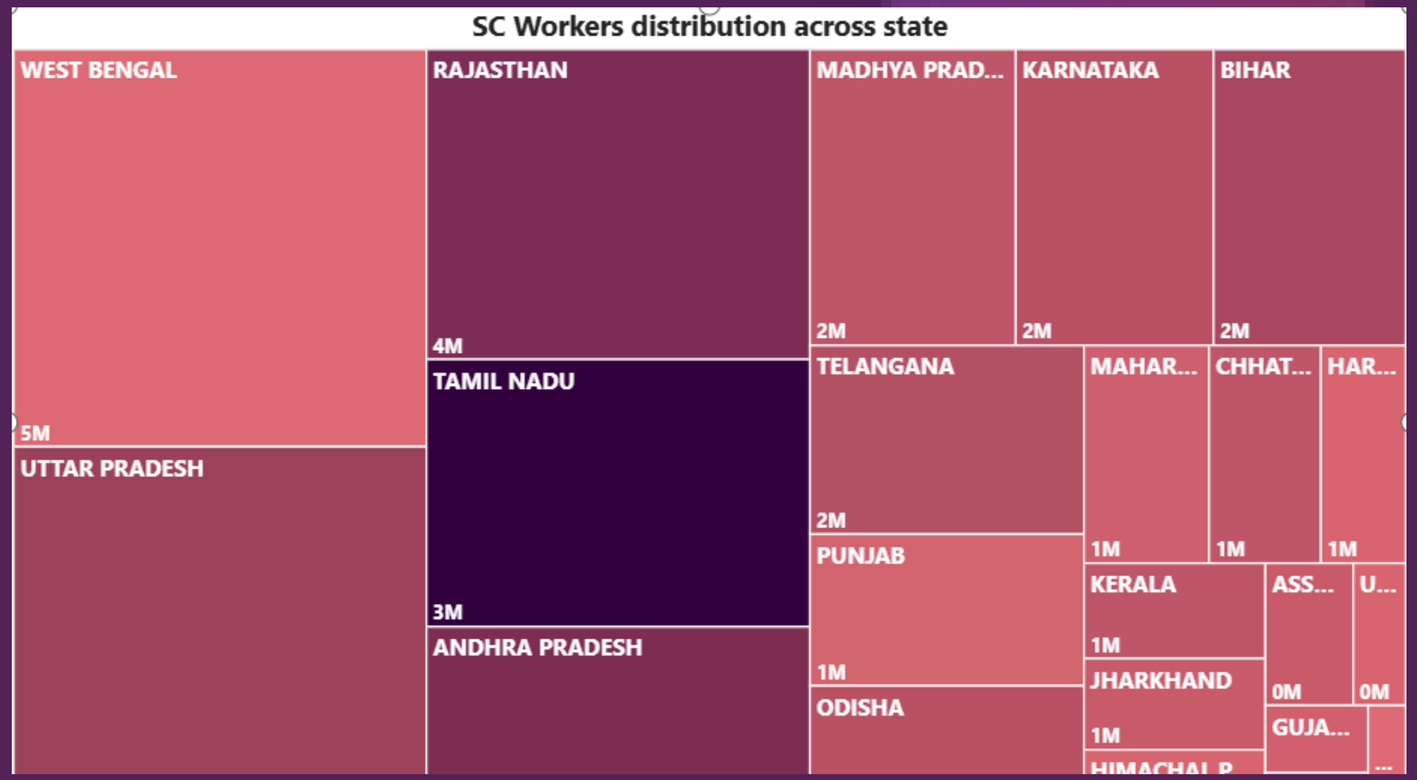
Works & Employent Analysis

QUESTION - 15

How does the number of SC persondays vary across states?

```sql
SELECT
    state_name,
    SUM(sc_persondays) AS total_sc_persondays
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_sc_persondays DESC;
```

state_name	total_sc_persondays
TAMIL NADU	67125407
UTTAR PRADESH	49144206
ANDHRA PRADESH	45843840
RAJASTHAN	33738219
BIHAR	20630227
TELANGANA	17085433
ODISHA	13286484
KARNATAKA	13227927
PUNJAB	12501159
MADHYA PRADESH	11626738



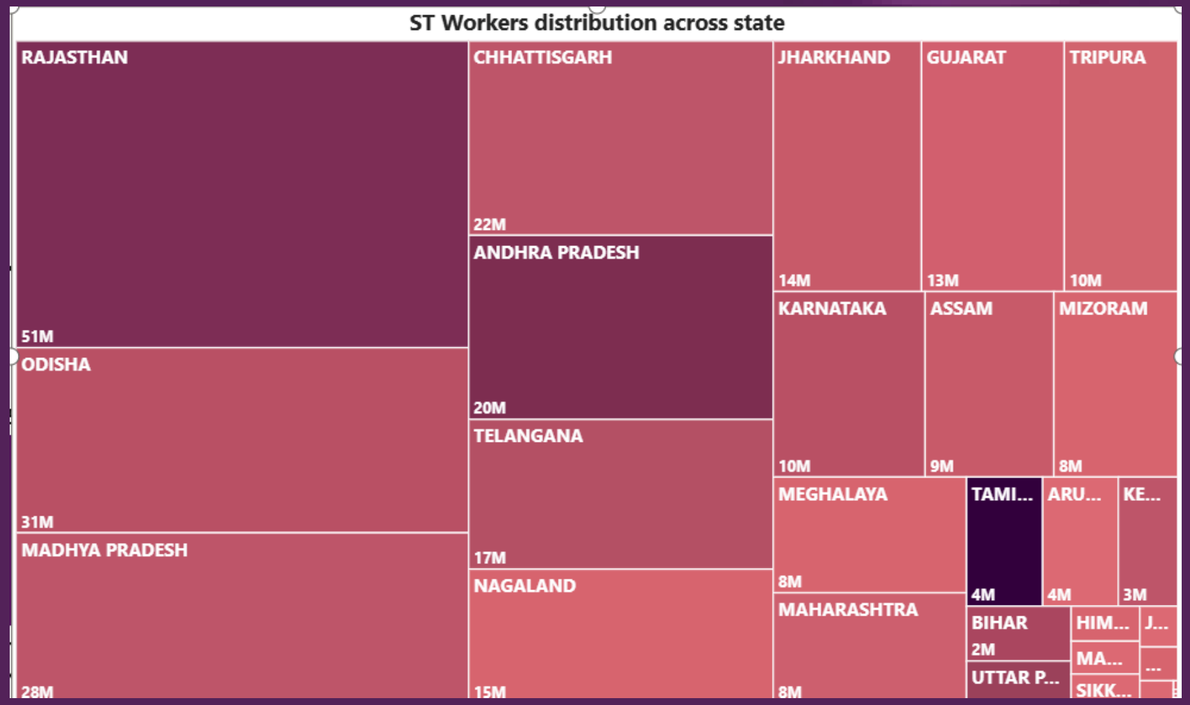
Works & Employent Analysis

QUESTION - 16

How does the number of ST persondays vary across states?

```sql
SELECT
    state_name,
    SUM(st_persondays) AS total_st_persondays
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_st_persondays DESC;
```

state_name	total_st_persondays
RAJASTHAN	50534895
ODISHA	30506197
MADHYA PRADESH	28010984
CHHATTISGARH	21558505
ANDHRA PRADESH	20411114
TELANGANA	16603232
NAGALAND	14831258
JHARKHAND	13537822
GUJARAT	13032073
TRIPURA	10351451



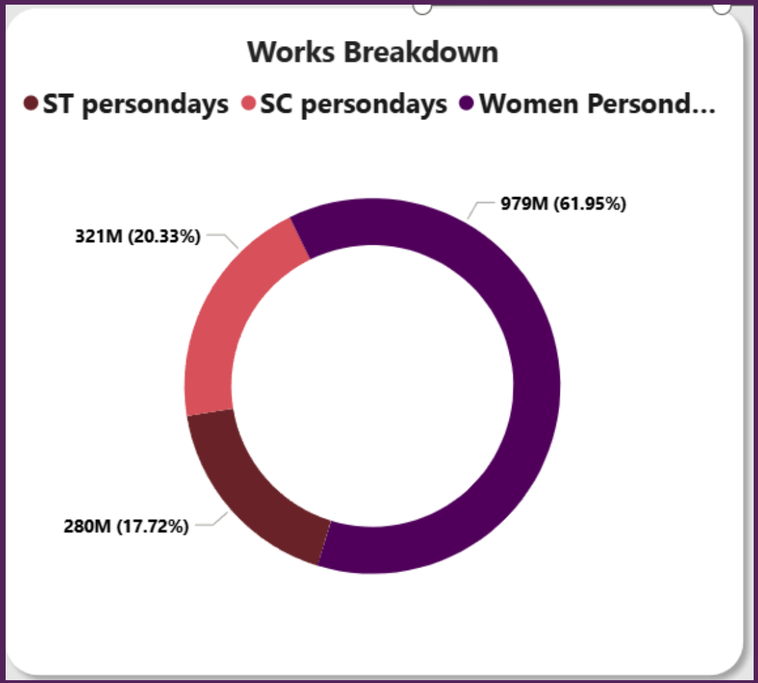
Works & Employent Analysis

QUESTION - 13

Workdays between Active social groups select sum(sc_persondays)

```
SELECT
    SUM(sc_persondays) AS sc_workdays,
    SUM(st_persondays) AS st_workdays,
    SUM(women_persondays) AS women_workdays
FROM mgnrega_data;
```

sc_workdays	st_workdays	women_workdays
321202295	279960190	978758223



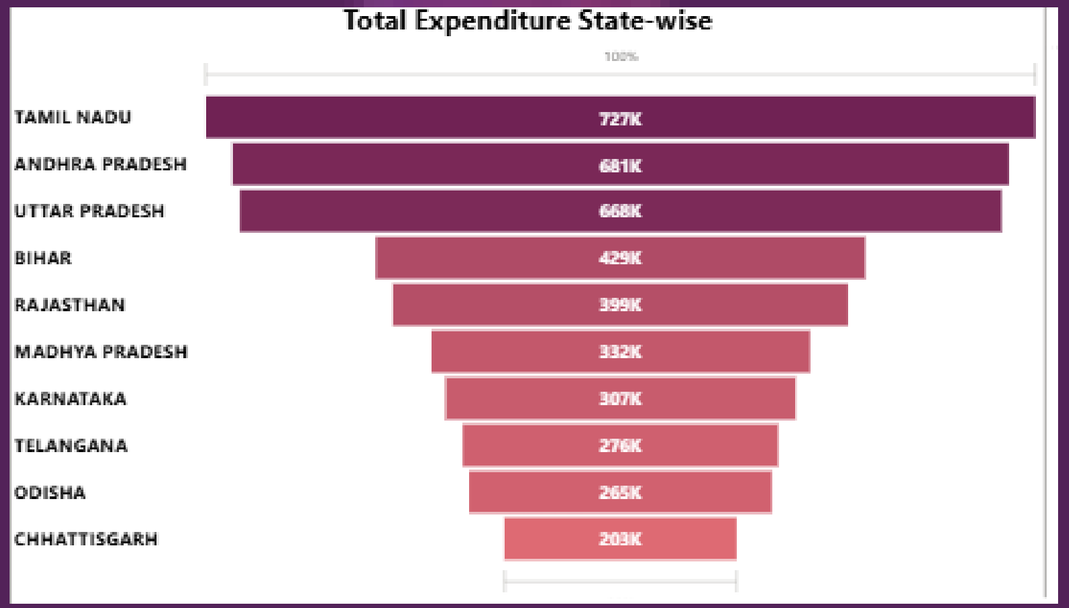
Budget & Expenditure

QUESTION - 18

Which states have the highest total expenditure?

```sql
SELECT TOP 5
    state_name,
    SUM(total_expenditure_lakhs) AS total_expenditure_lakhs
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_expenditure_lakhs DESC;
```

state_name	total_expenditure_lakhs
TAMIL NADU	727295.32
ANDHRA PRADESH	680813.71
UTTAR PRADESH	667834.82
BIHAR	428911.90
RAJASTHAN	399278.22



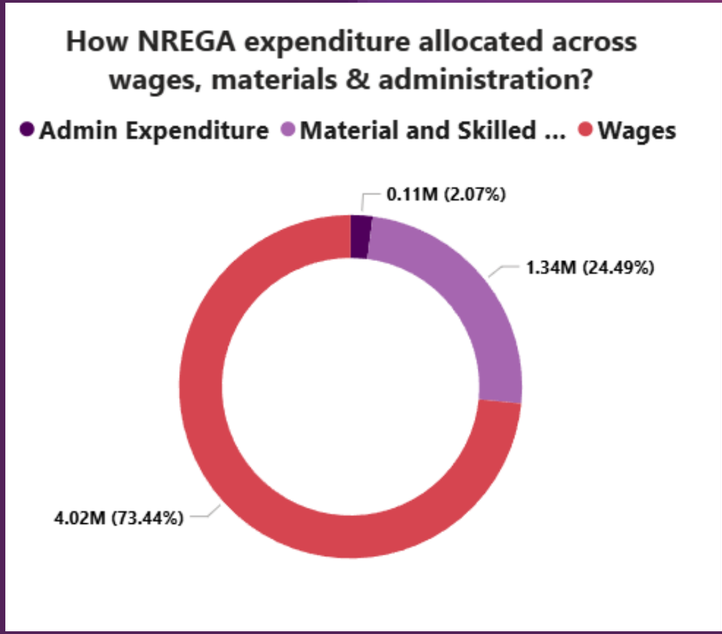
Budget & Expenditure

QUESTION - 19

What is the breakdown of expenditures on wages, material costs and admin expenditure?

```sql
SELECT
    SUM(wages_expenditure_lakhs) AS total_wages_expenditure,
    SUM(material_skilled_wages_lakhs) AS total_material_skilled_wages,
    SUM(admin_expenditure_lakhs) AS total_admin_expenditure
FROM mgnrega_data;
```

total_wages_expenditure	total_material_skilled_wages	total_admin_expenditure
4016983.60	1339728.97	113131.65



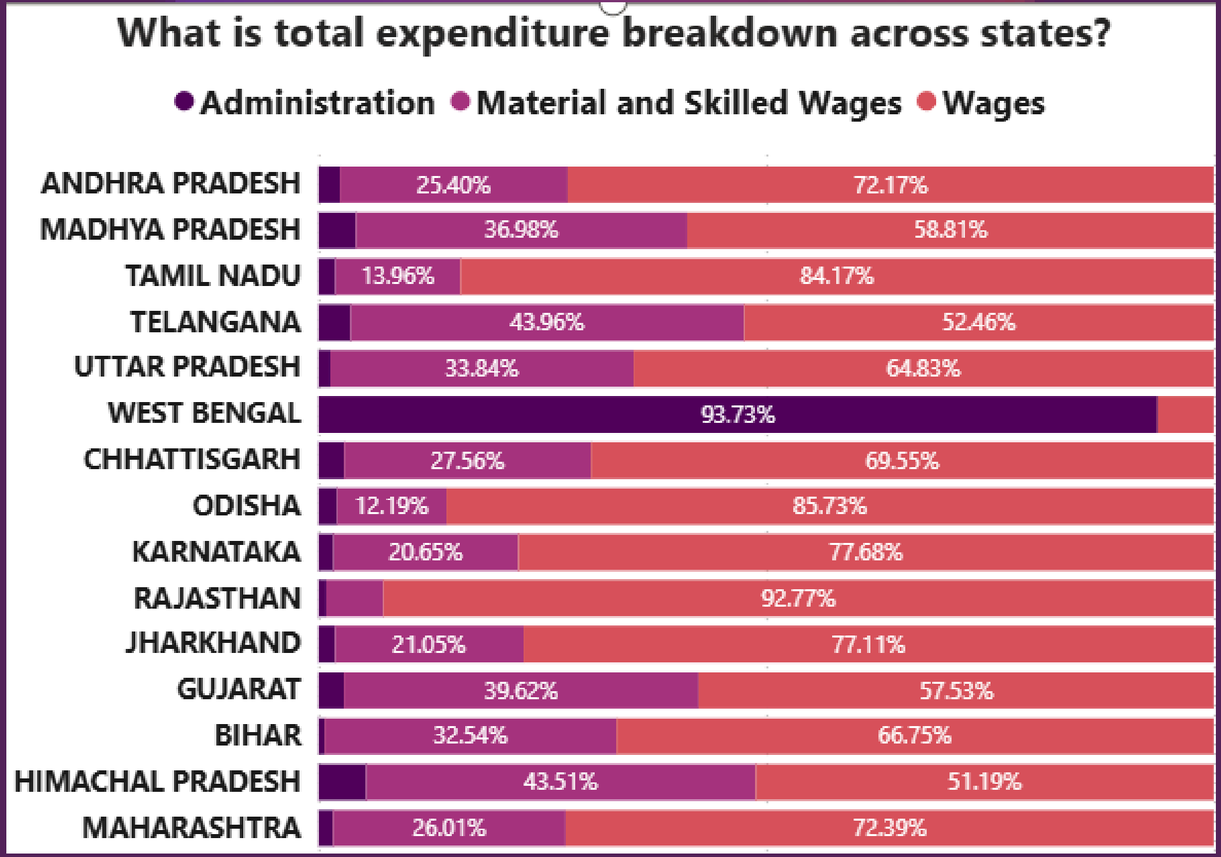
Budget & Expenditure

QUESTION - 20

What is the breakdown of expenditures on wages, material costs and admin expenditure in each state?

```sql
SELECT
    state_name,
    SUM(wages_expenditure_lakhs) AS total_wages_expenditure,
    SUM(material_skilled_wages_lakhs) AS total_material_skilled_wages,
    SUM(admin_expenditure_lakhs) AS total_admin_expenditure
FROM mgnrega_data
GROUP BY state_name;
```

state_name	total_wages_expenditure	total_material_skilled_wages	total_admin_expenditure
ANDAMAN AND NICOBAR	243.34	6.60	2.70
ANDHRA PRADESH	491338.99	172944.57	16530.18
ARUNACHAL PRADESH	9689.53	5583.21	259.03
ASSAM	124674.30	11533.21	0.00
BIHAR	286304.65	139579.47	3027.74
CHHATTISGARH	141523.29	56086.27	5866.62
DN HAVELI AND DD	0.00	0.00	0.00
GOA	73.38	59.30	1.80
GUJARAT	63247.94	43556.75	3137.64
HARYANA	18862.68	3698.06	550.84



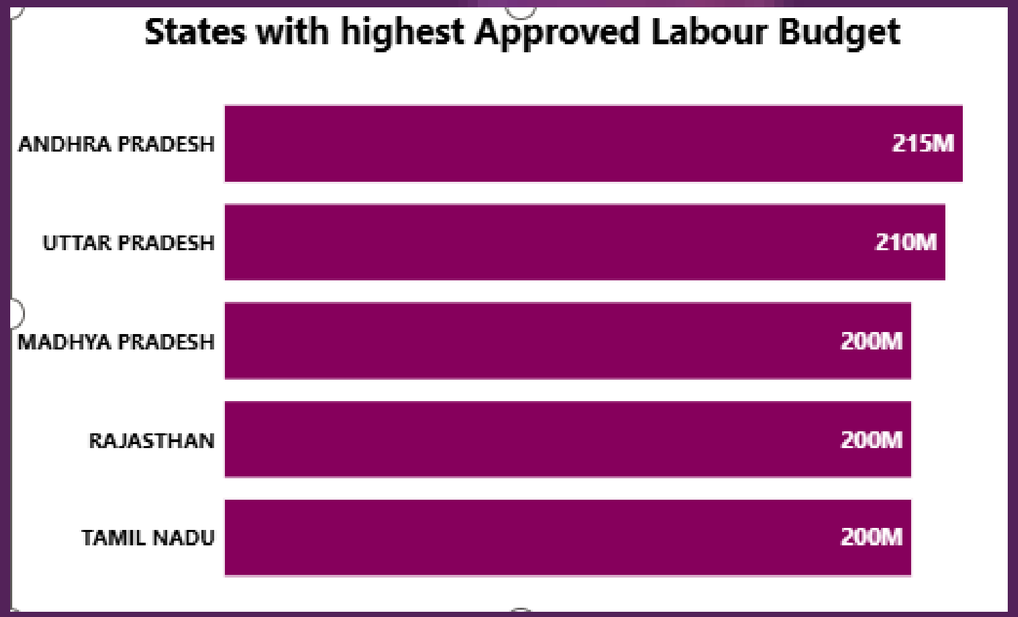
Budget & Expenditure

QUESTION - 21

Which states have the highest approved labor budget?

```sql
SELECT TOP 5
    state_name,
    SUM(approved_labour_budget) AS total_approved_labour_budget
FROM mgnrega_data
GROUP BY state_name
ORDER BY total_approved_labour_budget DESC;
```

state_name	total_approved_labour_budget
ANDHRA PRADESH	215000000
UTTAR PRADESH	210000000
MADHYA PRADESH	200000000
RAJASTHAN	200000000
TAMIL NADU	200000000



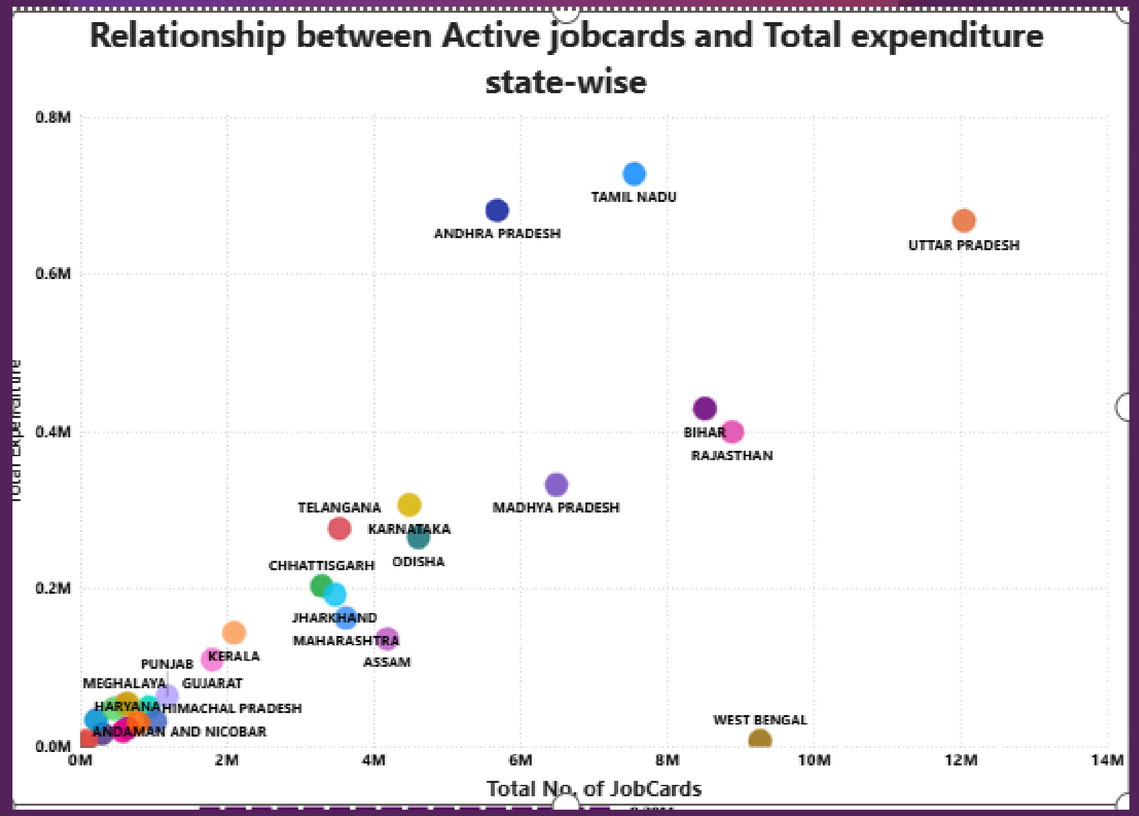
Budget & Expenditure

QUESTION - 22

Is there a relationship between active job cards and total expenditure?

```sql
SELECT
    state_name,
    SUM(total_active_jobcards) AS total_active_jobcards,
    SUM(total_expenditure_lakhs) AS total_expenditure_lakhs
FROM mgnrega_data
GROUP BY state_name;
```

state_name	total_active_jobcards	total_expenditure_lakhs
ANDAMAN AND NICOBAR	10904	252.64
ANDHRA PRADESH	5680334	680813.71
ARUNACHAL PRADESH	294308	15531.75
ASSAM	4184027	136207.50
BIHAR	8511127	428911.90
CHHATTISGARH	3290052	203476.21
DN HAVELI AND DD	1	0.00
GOA	7311	134.49
GUJARAT	1797977	109942.36
HARYANA	639828	23111.62



THANK YOU